Quantum Simulation of the Fermi-Hubbard Model with Ultracold Atomic Gases

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1 Abstract

Ultracold atomic gases in optical lattices offer a highly tunable platform for quantum simulation of the Fermi–Hubbard model, a paradigmatic system for strongly correlated electrons. Here, I review three landmark experiments: (i) direct imaging of the Fermi surface and Brillouin-zone filling in a noninteracting lattice gas, (ii) suppression of double-occupancy and characterization of the Mott insulator (iii) observation of divergent spin correlations, indicating the antiferromagnetic transition. Together, these studies validate key theoretical predictions, benchmark numerical methods in two and three dimensions, and demonstrate control over Hubbard parameters t, U, and chemical potential in attempt to explore phase diagram of the Fermi-Hubbard model.

2 Introduction

Since the turn of the century, ultracold quantum gases have emerged as a powerful platform for simulating condensed matter systems. By leveraging control techniques of atoms and lasers developed by atomic physicists throughout the 20th century, researchers can use precisely controlled ultracold atoms to realize the same model Hamiltonians that govern solid-state materials. In conventional solids, parameters such as interaction strength, potential landscape, and dimensionality are approximatingly fixed by the material itself. In

contrast, quantum simulation with ultracold gases offers a much larger scale of tunability of these parameters, allowing physicists to explore broad regions of the phase diagram under precisely controlled conditions. Quantum degenerate gases of fermionic isotopes, such as ^{40}K and ^{6}Li, serve as an effective platform for studying electrons in solids because they obey Fermi-Dirac statistics. The first realization of the quantum degenerate gas of fermions was achieved in 1999 using an evaporative cooling strategy to cool magnetically trapped ^{40}K atoms to $0.5T_F$ [1]. Following this, experiments turned to strongly interacting regimes, observing superfluidity across the BEC–BCS crossover in trapped Fermi gases [2]. Since those pioneering studies, many other alkali and alkaline-earth fermionic species have been cooled to quantum degeneracy for quantum simulation experiments.

In this paper, I focus on the quantum simulation of the Fermi–Hubbard model (FHM) using ultracold atomic gases trapped in optical lattices. The FHM serves as a cornerstone for understanding strongly correlated electrons in solids. Optical lattices are formed by intersecting laser beams that generate a spatially periodic trapping potential for atoms, thereby reproducing the crystalline structure of real solids—without phonons or defects—and directly realizing the FHM Hamiltonian. The FHM Hamiltonian, though deceptively simple, admits an exact analytical solution only in the infinite one-dimensional limit. In two and three dimensions, solving the model becomes numerically demanding, typically requiring advanced techniques such as dynamical mean-field theory and quantum Monte Carlo. This computational complexity makes the FHM particularly well-suited for exploration via quantum simulation.

This paper is organized as follows. First, we present a concise review of the fundamentals of many-body physics and control techniques in ultra-cold atomic systems [3] and the condensed matter principles underlying the FHM [4]. This overview is written at a level just sufficient to frame the subsequent experimental discussion. I then examine three seminal quantum-simulation experiments of the FHM: the first observation of the Fermi surface [5], the first realization of the Mott insulating phase [6], and the first observation of the

antiferromagnetic phase transition[7].

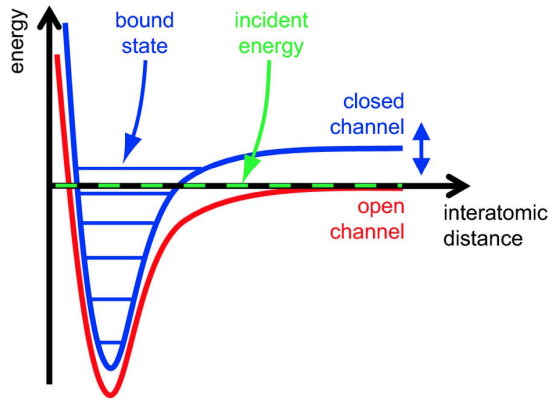
3 Background

3.1 Many Body Physics of Ultracold Gases

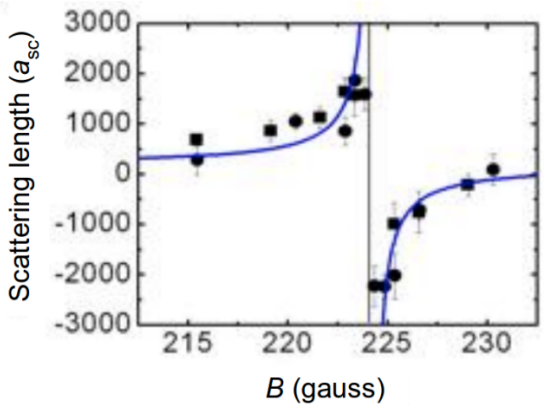
In order to observe quantum-statistical behavior—Bose–Einstein or Fermi–Dirac—in an atomic gas, the thermal de Broglie wavelength $\lambda_{dB} = h/\sqrt{2\pi m k_B T}$ must become comparable to the mean interparticle spacing $n^{-1/3}$. Equivalently, this occurs when $T \approx 0.2\,T_F$, giving typical degeneracy temperatures in the range of 1nK - 500nK [4]. At these ultralow temperatures, evaporative cooling is the method of choice: one continuously removes the highest-energy atoms from the trap and allows the remaining sample to rethermalize via elastic collisions. In a magnetic trap, atoms that explore regions of stronger field—hence have higher total energy—are driven by radiofrequency or microwave radiation into an untrapped hyperfine state and escape. In an optical dipole trap, lowering the trapping-beam intensity reduces the trap depth so that only the hottest atoms spill out. However, below roughly 1μK, only s-wave collisions remain energetically allowed—and s-wave scattering is forbidden between identical fermions by the Pauli exclusion principle. To evaporatively cool fermions one therefore employs sympathetic cooling, co-trapping them with a species (or spin state) that permits s-wave collisions. The "refrigerant" gas is cooled evaporatively, and the fermions thermalize to the same temperature by elastic collisions. A common example is cooling ^{40}K atoms via sympathetic evaporation with a ^{87}Rb boson gas [1, 5, 6].

At ultralow temperatures, only s-wave scattering contributes, and the two-body interaction is characterized by the scattering length a_{sc}. Its magnitude determines the strength of the interaction, while its sign indicates whether the interaction is attractive ($a_{\mathrm{sc}} < 0$) or repulsive ($a_{\mathrm{sc}} > 0$). We can tune a_{sc} over a wide range of vlaues using Feshbach resonances techniques [3], which exploit a resonant coupling between an open-channel scattering state and a closed-channel bound state. In a two-spin-component Fermi gas, the open and closed

channels may correspond to, for example, the triplet and singlet spin configurations, respectively. At short interatomic separations the hyperfine and exchange interaction provides an off-diagonal coupling between these channels. An external homogenous magnetic field shifts their relative energies via the Zeeman effect, and when the bound state is brought into near resonance with the scattering continuum the resulting hybridization causes the scattering length to diverge and change sign. By sweeping the magnitude of the magnetic field through the resonance field, one can therefore continuously tune the interaction from strengthening attractive ($a_{sc} < 0$), through the unitary limit ($|a_{sc}| \to \infty$), to weakening repulsive ($a_{sc} > 0$).



(a) Schematic of the two-channel model for a Feshbach resonance. Atoms initially collide in the open channel potential $V_{\rm op}(r)$ (red curve). During the collision, the open channel is coupled to a closed channel potential $V_{\rm cl}(r)$ (blue curve) that supports a bound molecular state. [3]



(b) Feshbach control of interactions. Scattering length $a_{\rm sc}$ as a function of magnetic field B for collisions between ^{40}K atoms in the $|F = 9/2, m_F = -9/2\rangle$ and $|F = 9/2, m_F = -5/2\rangle$ states. The Feshbach resonance appears as a sign-changing divergence in $a_{\rm sc}$, enabling tuning from attractive ($a_{\rm sc} < 0$) through effectively non-interacting ($a_{\rm sc} \approx 0$) to repulsive ($a_{\rm sc} > 0$) interactions.[4]

Figure 1: Feshbach Resonance.

To trap ultracold atomic gases in periodic potentials, one uses optical dipole traps to create an optical lattice. A single dipole trap exploits the dipole-induced energy shift caused by the inhomogeneous AC Stark effect. The atomic dipole interacts with the laser by creating an optical potential $V(\mathbf{r}) = d^2 E(\mathbf{r})^2/4\hbar\delta$, where d is the dipole matrix element, $\delta = \omega_L - \omega_0$ is the detuning of the laser frequency ω_L from the atomic resonance ω_0, and $E(\mathbf{r})$ is the

local electric-field amplitude. Since the intensity $I(\mathbf{r}) \propto E(\mathbf{r})^2$, a Gaussian beam in the z–direction produces an approximately harmonic confinement around its intensity maximum. For $\delta > 0$ (blue detuning) the potential is repulsive at the beam center, while for $\delta < 0$ (red detuning) it is attractive. To create a one-dimensional lattice along z, one overlaps two counterpropagating beams to form a standing wave $E(z) = E_0 \sin(k_L z)$. Repeating this along all three Cartesian axes yields a three-dimensional periodic lattice with potential [3]:

$$V(\mathbf{r}) = V_0 \Big[\sin^2(k_L x) + \sin^2(k_L y) + \sin^2(k_L z) \Big], \tag{1}$$

where $V_0 \propto d^2 E_0^2/(\hbar\,\delta)$ is the lattice depth and the lattice spacing in the resulting cubic array is $a = \pi/k_L$.

3.2 Fermi-Hubbard Model

The FHM in solid-state systems describes the dynamics of correlated electrons in a crystal lattice. The physics is governed by two main processes. First, electrons can tunnel between neighboring lattice sites, with the tunneling amplitude determined by the spatial overlap of the electronic orbitals localized at each site. Second, electrons interact via the Coulomb repulsion. In the standard (non-extended) FHM, this interaction is approximated by an on-site two-body repulsion: it acts only when two electrons occupy the same lattice site. For electrons in the lowest band, the Pauli exclusion principle restricts this to at most two electrons per site—one with spin up and one with spin down. The Hamiltonian is given by [4]:

$$\hat{H} = -t \sum_{\langle j,\ell\rangle,\sigma} \hat{c}^\dagger_{j\sigma} \hat{c}_{\ell\sigma} + h.c. + U \sum_j \hat{n}_{j\uparrow} \hat{n}_{j\downarrow} \tag{2}$$

where the first and second terms are the tunneling and interaction terms respectively. The first term consists of the creation and anhillation operators of electrons of spin σ on site j, denoted by $\hat{c}^\dagger_{j\sigma}$ and $\hat{c}_{j\sigma}$. The sum of $\langle j, \ell\rangle$ restricts the tunneling to and from neighboring sites. The tunelling strength t is given by the Hamiltonian single particle matrix element

for neighboring sites: $t = -\int d^3\mathbf{r}\ w^*_{\mathbf{R}_j}(\mathbf{r}) \left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{lattice}}(\mathbf{r})\right] w_{\mathbf{R}_\ell}(\mathbf{r})$, where $w(\mathbf{r} - \mathbf{R}_j)$ are the Wannier functions of the lowest band localized at the site at position $\mathbf{R}_\ell$. For the second interaction term, the single sum over j reflects only on-site coloumb interactions and the nummber operator $\hat{n}_{j\uparrow} = \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow}$. The strength of the interaction U for electrons in a solid is governed by the two-body Coulomb interaction at the same site, so $U = \int d^3\mathbf{r} \int d^3\mathbf{r}' w^*(\mathbf{r})w^*(\mathbf{r}')\mathbf{r}'w(\mathbf{r})w(\mathbf{r}')e^2/|\mathbf{r} - \mathbf{r}'|$ [8].

The assumptions and approximations of the FHM can be summarized as follows: (i) projection onto a single tight-binding band via Wannier functions; (ii) tunneling restricted to nearest neighbors; (iii) interactions limited to on-site terms; and (iv) absence of phonon coupling and lattice defects.

Despite these simplifications, the model captures the core physics of strongly correlated systems. The parameters U and t encode the competition between interaction and kinetic energy. In real solid-state materials, accurately determining and potentially tuning these parameters is a heavily studied task. Their ratio U/t controls the degree of correlation: for $U/t \gg 1$ the system is strongly correlated, while for $U/t \ll 1$ it is weakly correlated

3.3 Realization of the Fermi-Hubbarad Model in Ultracold Atomic Gases

Our current goal is to map the FHM Hamiltonian in Equation (2), which describes the dynamics of electrons on a crystal lattice, onto an analogous Hamiltonian describing ultracold fermionic gases trapped in an optical lattice. The first step is to encode the spin-up and spin-down states of electrons into a two-component Fermi gas. A common approach involves the fermionic isotope of ^{40}K. Being an alkali metal, ^{40}K has a single valence electron in the $4^2S_{1/2}$ ground state with nuclear spin $I = 4$. Therefore, in the absence of an external magnetic field, the ground state forms a hyperfine manifold of degeneracy $2F + 1$, where $F = 9/2$. This degeneracy can be lifted by the Zeeman effect in the presence of an external magnetic field and the lowest energy state $|F = 9/2, m_F = -9/2\rangle$ can then be prepared using

optical pumping. One can then apply an RF pulse to couple this state to another, such as $|F = 9/2, m_F = -7/2\rangle$ and create a controlled balanced mixture. These two hyperfine states can then represent our pseudospin $\sigma \in \{\uparrow, \downarrow\}$.

Once the two spin components are defined, prepared, and loaded into the optical lattice, the kinetic term in Equation (2) becomes directly analogous. The Wannier functions depend only on the tunneling particle's kinetic energy and the lattice potential V_{lattice}, which we substitute from Equation (1). The interaction term, however, needs to be slightly altered beacuse the interaction is no longer a Couloomb interaction between electrons but rather a contact interaction via scattering. The form of this interaction potential is of form $g\delta^3(\mathbf{r}-\mathbf{r}')$ where $g = 4\pi\hbar^2 a_{sc}/m$ [3]. Therefore, the interaction strength will be $U = g\int d^3\mathbf{r}|w(\mathbf{r})|^4$ and the FHM Hamiltonian is fully realized in our trapped fermionic gas.

The four assumptions of the FHM summarized in Section 3.2 can be realized in our ultracold-atom system. Assumption (i), projecting onto the lowest Bloch band, holds when both U and $k_B T$ remain below the energy gap to the first excited band. Furthermore, sufficiently deep lattice potentials guarantee that tunneling is restricted to nearest neighbors (ii) and that interactions occur only on-site (iii). Finally, the near-perfect periodicity of the optical lattice suppresses phonon coupling and lattice defects (iv).

Now we aim approximately solve for U and t in terms experimental parameters we can control. By solving the one-dimensional Mathieu equation in the deep-lattice limit $V_0 \gg E_r$ and restricting to the lowest Bloch band, one finds [3]:

$$\frac{t}{E_r} \approx \frac{4}{\sqrt{\pi}} \left(\frac{V_0}{E_r}\right)^{3/4} \exp\left(-2\sqrt{\frac{V_0}{E_r}}\right) . \tag{3}$$

where $E_r = \frac{\hbar^2 k_L^2}{2m}$ is the recoil energy and natural energy scale. Furthermore, approximating each lattice well by a harmonic potential and using the resulting ground-state Gaussian for

the Wannier function in the onsite-interaction integral yields [3]:

$$\frac{U}{E_r} \approx \frac{8}{\sqrt{\pi}} k_L \, a_{\mathrm{sc}} \left(\frac{V_0}{E_r} \right)^{3/4} \tag{4}$$

It can be seen from Equations (3) and (4) that the ratio U/t can be controlled by adjusting the lattice depth V_0, and it decreases exponentially with increasing V_0. Additionally, the interaction strength U can be tuned independently by varying the scattering length a_{sc} via a Feshbach resonance. Other parameters, such as the number of atoms in the trap and temperature, can be controlled by altering the evaporative cooling technique.

4 Experiments

4.1 Observation of the Fermi Surface

In one of the seminal experiments on fermions trapped in an optical lattice, researchers at Zurich in 2005 realized the single-band FHM and observed both the Fermi sea and the filling of the first Brillouin zone [5]. The experimental procedure closely followed the methods described in Section 3.3. First, ^{40}K atoms were optically pumped into the $|F = 9/2, m_F = -9/2\rangle$ state and sympathetically cooled with ^{87}Rb atoms via microwave radiation in a magneto-optical trap. The ^{40}K atoms were transferred to an optical trap, where a spin mixture with a population of $(50 \pm 4)\%$ was prepared in each of the two spin states $|F = 9/2, m_F = -9/2\rangle$ and $|F = 9/2, m_F = -7/2\rangle$. Once evaporativly cooled to quantum degeneracy at approximately $T/T_F = 0.2$, using Feshbach resonance techniques, the s-wave scattering length between the two spin species was tuned to zero, effectively rendering the two spin components non-interacting ($U = 0$). Subsequently, an optical lattice with a simple cubic structure was formed to realize the desired system.

The parameter that was varied in this experiment to observe the Fermi sea is the characteristic density $\rho_c = \frac{Nd^3}{\zeta_x \zeta_y \zeta_z}$, where d is the lattice spacing and $\zeta_\alpha = \sqrt{2t/m\omega_\alpha}$ is the

characteristic length scale along the α direction. The characteristic length ζ_α defines the distance over which the tunneling strength t equals the confinement energy of the harmonic potential, and thus the product $\zeta_x\zeta_y\zeta_z$ effectively sets the trap volume. The characteristic density ρ_c then represents the number of atoms per trap volume, in units of atoms per lattice site. This quantity is of interest and varied because, at $T = 0$, the quasi-momentum distribution of atoms in the lattice depends only on ρ_c. As ρ_c increases, more of first Brillouin zone becomes occupied, or equivalently, the Fermi energy increases. Experimentally, ρ_c can be adjusted either by increasing the trap depth V_0, which decreases the tunneling amplitude t, or by varying the number of atoms N loaded into the trap.

In this experiment, the population within the Fermi sea was mapped out by adiabatically decreasing the lattice potential to zero so that atoms remain in the lowest Bloch band with quasi-momentum conserved. Once the lattice potential vanishes, the atoms underwent 9ms of ballistic (free) expansion, mapping their in-lattice quasi-momentum distribution—determined solely by the characteristic density ρ_c—onto real-space positions. The resulting absorption images in Figur 2a) reveal that at low ρ_c the occupied quasi-momentum states lie near $\mathbf{k} = 0$, where the tight-binding dispersion is approximately quadratic, yielding a circular Fermi sea. As ρ_c increases, the Fermi energy approaches the Brillouin-zone boundary at $|k_x|, |k_y| \leq \pi/d$, and the occupied region deforms into the square defined by the first Brillouin zone. Observing this square thus directly images the Fermi sea of the fully filled tight-binding model.

4.2 Observation of Fermionic Mott Insulator

While in the previous experiment ideal non-interacting conditions were used to observe the Fermi surface, researchers again at Zurich in 2008 enter the strongly correlated regime of the FHM to observe and study Mott insulators [6]. A Mott insulator is a phase in which the lowest Bloch band is partially filled, yet strong on-site repulsion U localizes one fermion per site with spin $\uparrow$ or $\downarrow$, producing insulating behavior. The competition between the

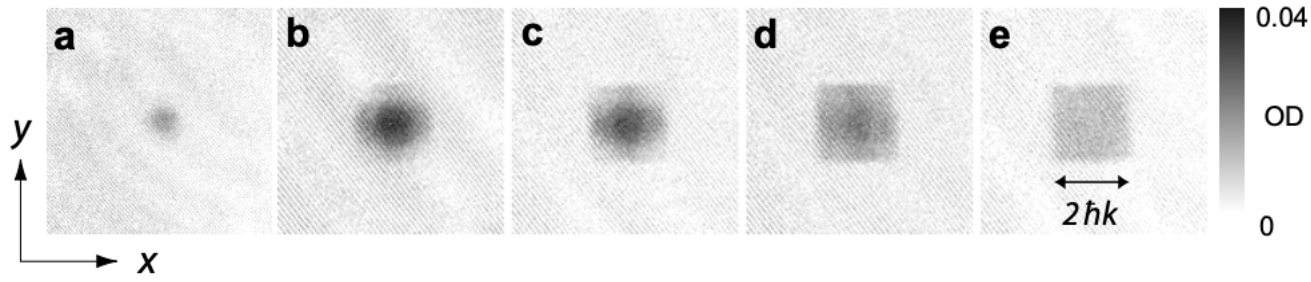


Figure 2: Observing the Fermi Surface. Images of ultracold fermions after adiabatically ramping down the optical lattice and 9 ms of ballistic expansion, showing the emergence and deformation of the Fermi surface as the characteristic density and lattice depth vary. (a) 3,500 atoms per spin state at a lattice depth of $5\,E_r$. (b)–(e) 15,000 atoms per spin state with lattice depths of $5\,E_r$ (b), $6\,E_r$ (c), $8\,E_r$ (d), and $12\,E_r$ (e).[5]

interaction energy U, the tunneling amplitude t, and the trap Fermi energy $E_F \propto N_\sigma^{3/2}\,\omega^2$ (with $N_\sigma = N_\uparrow = N_\downarrow$) determines the phase: in the metallic regime $U \ll E_F < 12t$ the band is partially filled and interactions are weak; in the band-insulating regime $U \ll 12t < E_F$ the band is completely filled and the system is insulating due to band structure; whereas in the Mott-insulating regime $E_F < 12t \ll U$ strong correlations at half filling freeze transport [9].

In this experiment the researchers also employed a quantum-degenerate gas of ^{40}K atoms in which two sublevels of the $F = 9/2$ hyperfine manifold encode the spin-$\uparrow$ and spin-$\downarrow$ states. To observe Mott insulating behavior they measured the number of doubly occupied lattice sites, D, as a function of the total atom number N. In the metallic regime ($U \ll 12t$) the double-occupancy D grows rapidly with N as the lowest Bloch band fills two fermions per site, whereas in the Mott-insulating regime ($U \gg 12t$) strong on-site repulsion suppresses double occupancy ($D \approx 0$) even as N increases.

The following experimental procedure was executed.^{40}K were sympathetically cooled with ^{87}Rb and balanced mixture of N ^{40}K atoms in the $|m_F = -9/2\rangle$ and $|m_F = -5/2\rangle$ hyperfine states were loaded into an optical lattice. The on-site interaction U and the tunneling amplitude t were then set by tuning the magnetic bias field near the Feshbach resonance and

adjusting the lattice depth. To measure D for varying experimental parameters, tunneling was first frozen by raising the lattice depth to $V_0 = 30\,E_r$, and the magnetic field was tuned to a value where a bound molecular state exists for the $|-9/2,-7/2\rangle$ pair, while the $|-9/2,-5/2\rangle$ pair remained free. A radiofrequency π-pulse resonant with the molecular transition was then applied, selectively flipping $|-7/2\rangle \leftrightarrow |-5/2\rangle$ on doubly occupied sites. This process dissociated $|-9/2,-7/2\rangle$ molecules into free atoms or associated $|-9/2,-5/2\rangle$ atoms into molecules, while singly occupied sites remained unaffected.

Subsequently, the magnetic field was ramped down to dissociate any remaining molecules, the lattice was switched off, and a magnetic field gradient was applied to spatially separate the spin states. Since only $|-9/2\rangle$ and $|-5/2\rangle$ atoms were initially present in the optical lattice, the number of $|-7/2\rangle$ atoms observed in the resulting absorption image directly yielded the double occupancy D. The results from this experiment are displayed in Figure 3. In Figure 3a), a significant increase of D with N is observed in the noninteracting regime (empty circles), whereas upon entering the Mott insulating regime (filled circles), the double occupancy is suppressed. In Figure 3b), a comparison between different Mott regimes shows that D is suppressed more strongly for larger interactions, although a slight increase in D indicates the presence of a metallic region near the trap center.

4.3 Observations of the Anti-Ferromagnetic Phase

Since the development of quantum simulation of the FHM using ultracold gases in optical lattices, significant effort has been devoted to observing spin correlations and the antiferromagnetic phase transition. Such an observation would provide experimental access to phases fundamental to models of high-temperature superconductivity and quantum magnetism.

The antiferromagnetic phase of the FHM can be observed by performing second-order perturbation on the tunneling term ($U >> t$):

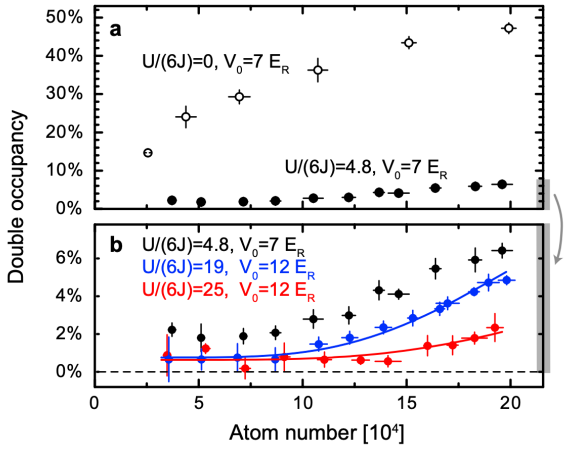


Figure 3: Double-occupancy fraction D vs. total atom number N in (a) the non-interacting and weakly interacting regime (empty circles) and the Mott insulating regime (filled circles). In (a), D rises steeply with N when $U \approx 0$, while strong repulsion suppresses D in the Mott insulator. (b) A zoom into the Mott regime shows D remains near zero until large N, where a slight increase indicates a metallic core in the trap center. Solid blue and red curves are atomic-limit theoretical predictions [6].

$$\hat{H}_{\mathrm{FHM}} \overset{\text{2nd–order PT}}{\longrightarrow} \hat{H}_{\mathrm{Heis}} = J \sum_{\langle j,\ell \rangle} \hat{\mathbf{S}}_j \cdot \hat{\mathbf{S}}_\ell \,, \quad J = \frac{4t^2}{U} \,. \tag{5}$$

The Heisenberg Hamiltonian captures antiferromagnetism when $J > 0$. A critical challenge in realizing the antiferromagnetic phase is that it emerges near the superexchange scale or the Néel temperature, $T_N = J/k_B = 4t^2/k_B U$, which in the strongly correlated regime ($U >> t$) corresponds to $0.02 - 0.05\,T_F$ —an order of magnitude colder than the temperatures required for achieving quantum degeneracy. Additionally, new diagnostic tools had to be developed to probe spin correlations at these extremely low temperatures.

Previous progress toward this goal included experiments that observed antiferromagnetic spin correlations over short [10] and exntended distances [11], but it was not until recently that a group of researchers in China observed the long-range antiferromagnetic order. This phase is characterized by the divergence of the spin structure factor (SSF) $S(\mathbf{q})$ and spontaneous SU(2) symmetry breaking. The spin structure factor is given by $S(\mathbf{q}) = \frac{1}{N} \sum_{i,j} e^{i\mathbf{q}\cdot(\mathbf{r}_i - \mathbf{r}_j)} \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle$, which measures spin-spin correlations in momentum space

across the entire lattice. Since antiferromagnetism corresponds to alternating spins, the relevant quantity is $S(\pi)$, where $\pi = (\pi, \pi, \pi)$ in a three-dimensional lattice. Due to spontaneous symmetry breaking, the spin structure factor exhibits divergent behavior near the Néel temperature, scaling as $S_\pi \propto (T - T_N)^{-\gamma}$, where $\gamma = 1.396$ is the critical exponent.

This experiment measures the expected critical divergent behavior of the SSF in a three-dimensional FHM system comprising ^{6}Li atoms in an optical lattice with approximately 800,000 sites, exhibitting strong evidence for the realization of the antiferromagnetic phase. The breif experimental scheme is as follows. To investigate long-range antiferromagnetic order in the 3D FHM, it is essential to overcome challenges present in earlier optical lattice experiments: the spatial nonuniformity of hopping t, interaction U, and chemical potential μ due to Gaussian laser beam profiles, and the heating caused by density redistribution during lattice loading. They addressed these issues by developing a novel technique that combines a flat-top optical lattice with a box trap, creating a highly uniform central region where t, U, and μ are essentially constant. ^{6}Li were cooled $0.041T_F$ and the atoms were prepared equally populated in the two hyperfine spin states $|F = 1/2, m_F = 1/2\rangle(\uparrow)$ and $|F = 3/2, m_F = -3/2\rangle(\downarrow)$ After adiabatically loading the gas into the flat lattice at near half-filling of the lowest band, they employed spin-sensitive coherent Bragg scattering, where spin-dependent phase shifts of scattered photons allowed direct measurement of the spin structure factor $S(\pi)$.

The results are displayed in Figure 5. Figure 5a) is a schematic phase diagram of the 3D FHM at half filling. The red dot represents the calculated temperature at $U/t = 0$, and the red curve indicates the temperature as a function of U/t during the experiments. Figure 5b) shows $S(\pi)$ versus U/t fitted by $S_\pi - S_0 \propto \left|\frac{U_c}{U} - 1\right|^{-\gamma}$, where S_0 is the background contribution and $\gamma = 1.396$ is fixed. The plot clearly shows critical divergences in the spin structure factor, which is "smoking-gun evidence" of the antiferromagnetic transition.

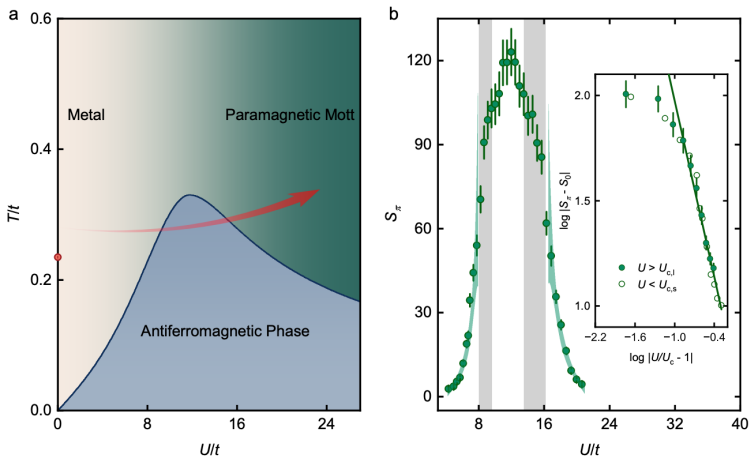


Figure 4: Measurement of spin structure factoor(a) Schematic 3D Fermi–Hubbard phase diagram at half filling. The red dot marks the estimated temperature at $U/t = 0$ after adiabatic lattice loading; the curved band shows its evolution with U/t, with deviations arising from slight nonadiabaticity. (b) Measured spin structure factor S_π vs. U/t. Power-law fits $S_\pi - S_0 \propto |U/U_c - 1|^{-\gamma}$ with $\gamma = 1.396$ are shown for $U/t < 8$ and $U/t > 16.4$; green and gray bands are 95% confidence intervals on S_π and U_c/t.[7]

5 Conclusion and Outlook

Quantum simulation of the Fermi–Hubbard model with ultracold atomic gases has, through the direct imaging of the Fermi surface, the measurement of double-occupancy in the Mott insulator, and the observation of divergent spin correlations in the antiferromagnet, provided an unprecedented, bottom-up validation of strongly correlated-electron physics. By realizing each landmark regime under tunable, defect-free conditions, these experiments confirm key theoretical predictions and benchmark numerical methods in two and three dimensions. Looking forward, advances in cooling—particularly improved entropy-redistribution schemes—could allow us to push below the critical temperature T_c for d-wave pairing and thus explore analogues of high-temperature superconductivity. Other promising directions include the investigation of transport properties in the Hubbard system and the study of its non-equilibrium dynamics.

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